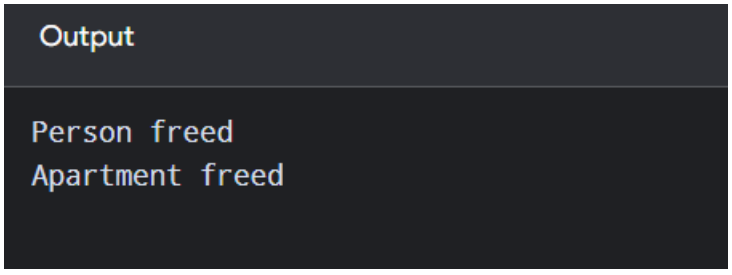
Task 7:

1.
```
class Person {
    var apartment: Apartment?
    deinit { print("Person freed") }
}
class Apartment {
    weak var tenant: Person?
    deinit { print("Apartment freed") }
}
var p: Person? = Person()
var a: Apartment? = Apartment()
p?.apartment = a
a?.tenant = p
p = nil
a = nil
```

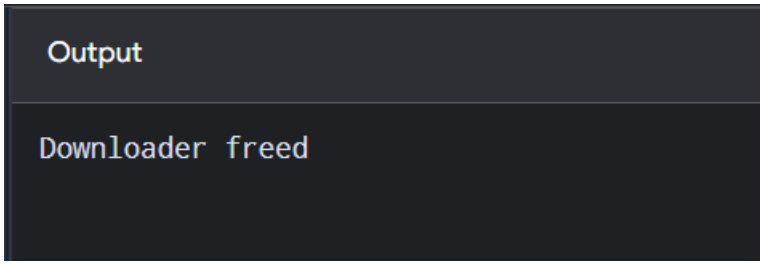


2.
```
class Downloader {
    var name: String
    var onFinish: (() -> Void)?
    init(name: String) {
        self.name = name
    }

    func start() {
        onFinish = { [weak self] in
            if let self = self {
                print("Finished \(self.name)")
            }
        }
    }

    deinit {
        print("Downloader freed")
    }
}

var d: Downloader? = Downloader(name: "File")
d?.start()
d = nil
```



3.

Output

```
before c
Box gone
end
```